

ContextPrior-PLM for Stability Prediction

1. Input & Frozen PLM

Protein Sequence
M S E K L R Q A ... L K K I E

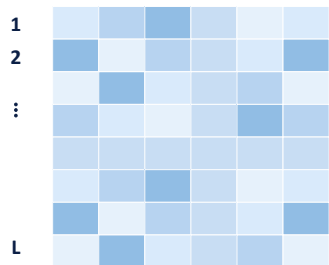
Length = L



Frozen PLM Encoder
(ESM2- 650M)



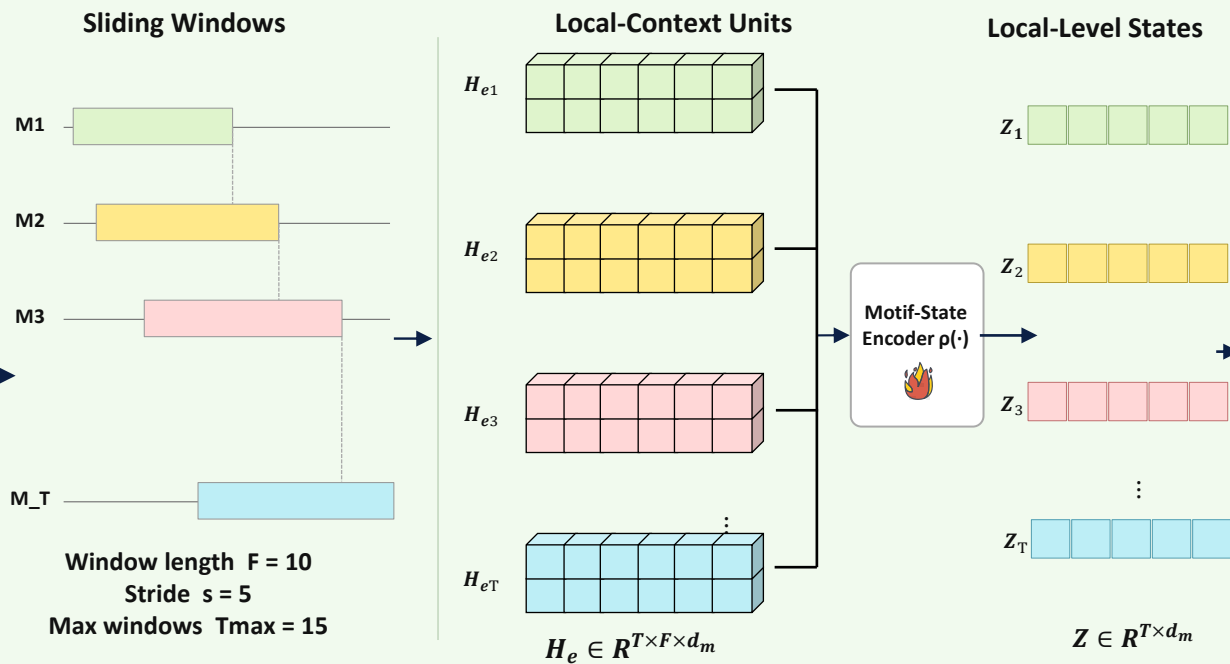
Residue-level States



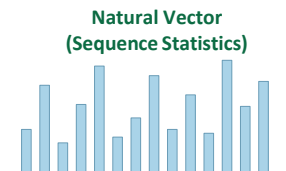
$$H\phi \in \mathbb{R}^{L \times d\phi}$$

- Frozen (no update)
- LOCK Fixed parameters
- Trainable
- Residual connection

2. Local-Context Construction



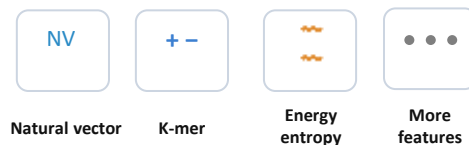
Compute Statistics Descriptors for Each Local-Context Unit



1 5 10 15 ... F

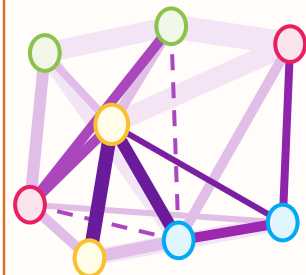
Amino acid occurrences
& positional statistics

Other Statistics



3. Context Prior Graph & Injection

Context Prior Graph



Edge weight (w^{ij})
Low High
Similarity in descriptor space
(not physical contacts)

Residual Graph Injection

Graph Operator

$$G(Z, A)$$

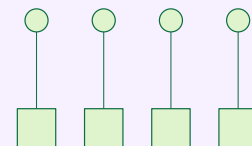
$$Z' = Z + G(Z, A)$$

$$Z' \in \mathbb{R}^{T \times d_m}$$



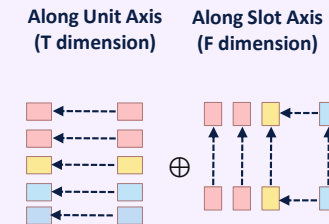
4. Axial Local-Context Encoder & Prediction

Local Graph Aggregation



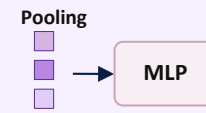
Intra-unit message passing
captures local dependencies

Axial Encoder (Dual Attention)



Captures interactions across units

Readout



Stability Score

